



FANTASTECH ENERGY SOLUTION LLC

KOOR Active Stack

51.2V Stackable LiFePO4 Battery — Spec Sheet

51.2V • 314Ah • 16 kWh

Model: KOOR-51.2-314AH-16KWH



4-section active stack

GRADE A CELLS

ACTIVE CELL BALANCING

AUTO PRECHARGE

BUILT-IN HEATER PADS

10-YEAR WARRANTY

ELECTRICAL SPECIFICATIONS

Nominal Voltage	51.2V
Nominal Capacity	314Ah
Nominal Energy	16.08 kWh
Cell Chemistry	LiFePO4 — Grade A cells, individually tested & matched (OCV/IR), laser-welded interconnects
Charging Voltage	56.0V – 58.4V
Discharge Cutoff Voltage	43.2V
Charge Method	CC-CV, standard 62.8A (0.2C), cutoff 6.28A (0.02C)
Max. Charge / Discharge Current	200A continuous / 200A continuous
Parallel Expansion	Up to 15 units, one comm. chain (auto-address, RS485/CAN cascade)

BATTERY MANAGEMENT & PROTECTION

Built-in BMS	Monitors & protects voltage, current, and temperature on every string
Cell Balancing	Active balancing, 2A balance current — actively moves charge between cells instead of just bleeding it off, keeping the pack in step over its life
Precharge Circuit	Factory-installed automated precharge — 30Ω/200W resistor with a 5-8 second time-delay relay. Soft-starts the inverter on its own; no manual precharge button
Protections	Over/under-voltage, overcurrent, short circuit, over/under-temperature, cell imbalance
Communication	Dual RS485, CAN, RS232

THERMAL MANAGEMENT

Heating	Internal heater pads wired into the BMS heating circuit — automatic sub-zero charge protection, no external heater needed
Charge Temp. Range	0°C to 45°C (32°F to 113°F)
Discharge Temp. Range	-20°C to 60°C (-4°F to 140°F)
Storage Temp. Range	-20°C to 45°C (-4°F to 113°F)

DISPLAY & MONITORING

Screen	4.3" TFT smart touch screen, built into front control panel
Live Readouts	Voltage, current, SOC, temperature
Status Indicators	RUN / ALM / capacity LEDs

MECHANICAL / PHYSICAL

Footprint (W x D)	720 x 526 mm (28.3 x 20.7 in)
Overall Height	≈1,176 mm / 46.3 in (base + 4 stacked sections + control head)
Weight	130 kg ± 5 kg (287 lb ± 11 lb)
Enclosure	Steel, floor-standing modular stack

CERTIFICATIONS & WARRANTY

Transport	UN38.3 tested, MSDS on file — per shipment
Safety Listing	Not UL Listed
Warranty	10-Year Manufacturer Warranty

WHAT SETS THE ACTIVE STACK APART

- **Active cell balancing** — keeps every cell matched over the pack's full life instead of just bleeding the highest cells down.
- **Automated precharge** — the pack soft-starts the inverter itself on power-up; no manual precharge switch to remember.
- **Built-in heater pads** — protects the pack from sub-zero charging without a separate space heater or heat tape.
- **Grade A cells** — every cell individually tested and matched before the pack is built.
- **10-Year Warranty**, backed by a manufacturing partner with 14 years in LiFePO4 production.

Specifications may change without prior notice. Not UL Listed — confirm code and AHJ requirements before specifying on a permitted job. © Fantastech Energy Solution LLC.